

Rajarata University of Sri Lanka

Department of ICT
ICT 1209 – Web Technologies

Mini Project – Final Report

TasteHub

Theme: Digital Recipe Book

Group Member 1	G Chathushi Prabodika – 2768
Group Member 2	P.R.Samadhi Asanthika– 2794
Date of Submission	[22/09/2026]
GitHub Repository	[https://github.com/Chathushi-Prabodika/TasteHub.git]

Table of Contents

1. Project Overview
2. System Design
3. Implementation
4. Challenges and Solutions
5. Individual Contribution
6. Setup Instructions 7. References

1. Project Overview

1.1 Purpose

TasteHub is a dynamic web application that solves the problem of finding, organizing, and sharing home-cooked meals by providing a centralized digital recipe book[cite: 8]. We chose this theme to connect culinary enthusiasts through a modern, responsive platform that makes recipe discovery effortless

1.2 Target Audience

The application is designed for home cooks, university students looking for quick meals, and culinary enthusiasts who need a reliable digital space to discover daily food inspiration and share their own signature recipes.

1.3 Key Features

- **User Authentication:** Secure registration and login system with encrypted passwords and session management.
- **Dynamic Recipe Management:** Users can view detailed recipes, categorize them, and submit their own creations with image uploads.
- **Interactive UI:** Seamless navigation with dynamic content updates, responsive Bootstrap 5 cards, and modern JavaScript form validations.
- **Community Support:** A direct contact portal storing user feedback and inquiries directly into the database.

2. System Design

2.1 Database Schema

Describe the tables in our MySQL database (*tastehub_db*).

Table	Field	Type	Description
users	id	INT (PK, AI)	Auto-increment primary key
	username	VARCHAR	User chosen Display Name
	email	VARCHAR	User Email address(unique)
	password	VARCHAR	Hashed with password_hash()
	created_at	TIMESTAMP	Account creation date and time
messages	id	INT(PK,AI)	Auto increment primary key
	name	VARCHAR(100)	Sender's full name
	email	VARCHAR(100)	Sender's email address
	message	TEXT	The contact from submission content
	created_at	TIMESTAMP	Time the message was submitted
Recipe	id	INT(PK,AI)	Auto increment primary key
	title	VARCHAR(100)	Name of the recipe

	category	VARCHAR(100)	categories
	ingredients	TEXT	List of recipe ingredient
	instructions	TEXT	Step by step cooking instructions
	image	VARCHAR(225)	File path to the uploaded image
	user_id	INT(FK)	Foreign key linking to users table

2.2 Folder Structure

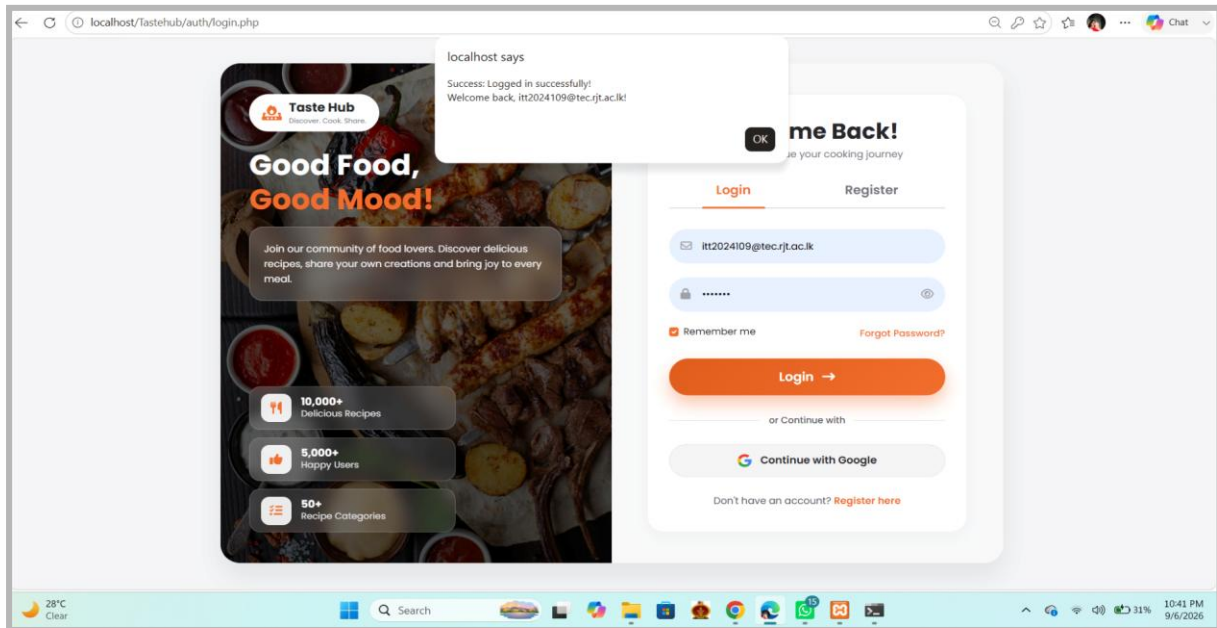
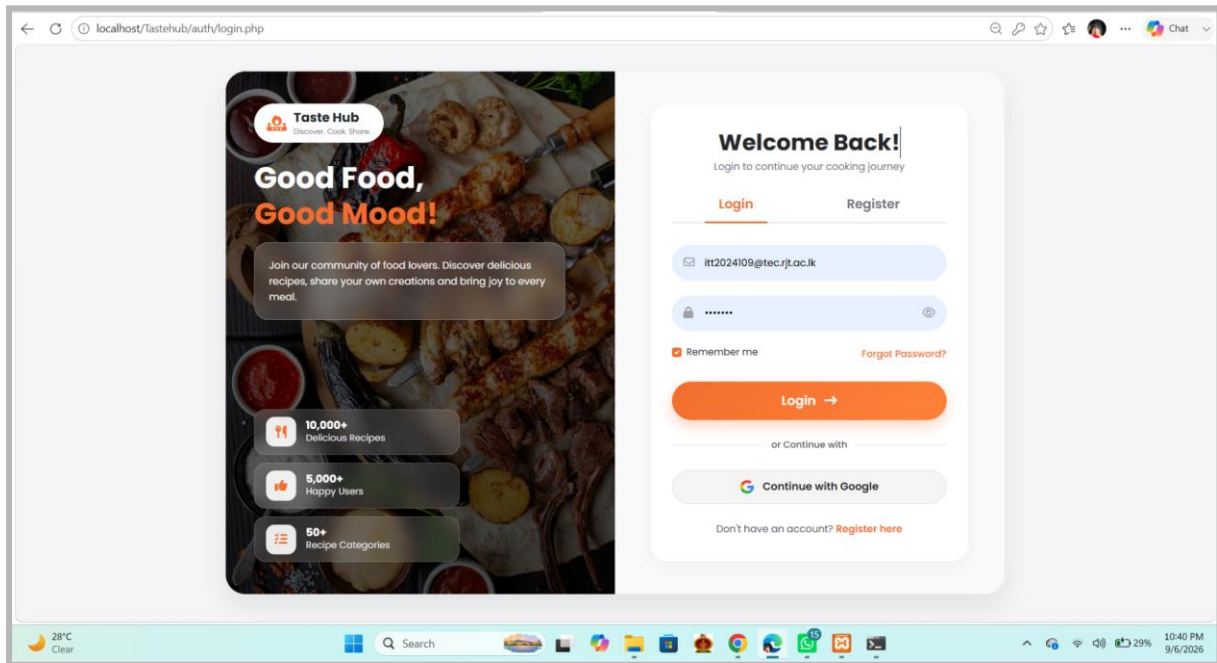
- TasteHub/
 - |— css/
 - | |— style.css
 - | |— login.css
 - |— js/
 - | |— main.js
 - | |— recipe.js
 - |— images/
 - | |— hero banner.png.png
 - | |— garlic creamy pasta.jfif
 - | |— [uploaded recipe images]
 - |— includes/
 - | |— db.php
 - | |— functions.php
 - |— auth/
 - | |— login.php
 - | |— register.php
 - | |— logout.php
 - |— contact.php
 - |— dashboard.php
 - |— index.php
 - |— recipes.php
 - |— recipe-details.php
 - |— database.sql
 - |— README.md

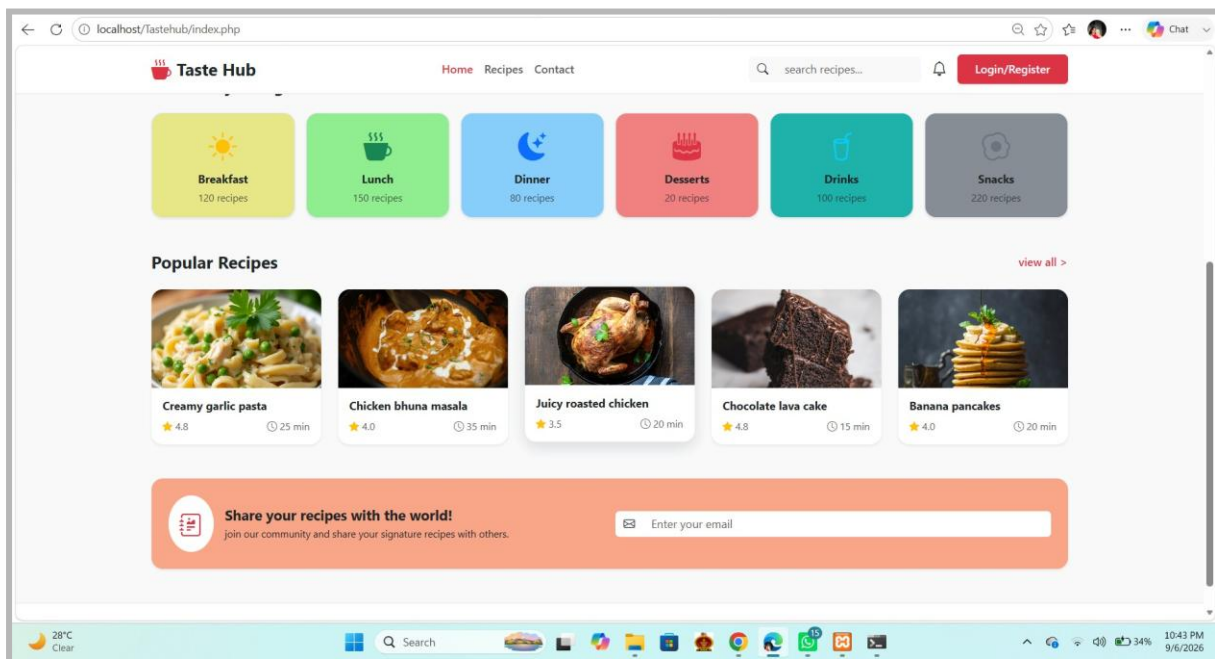
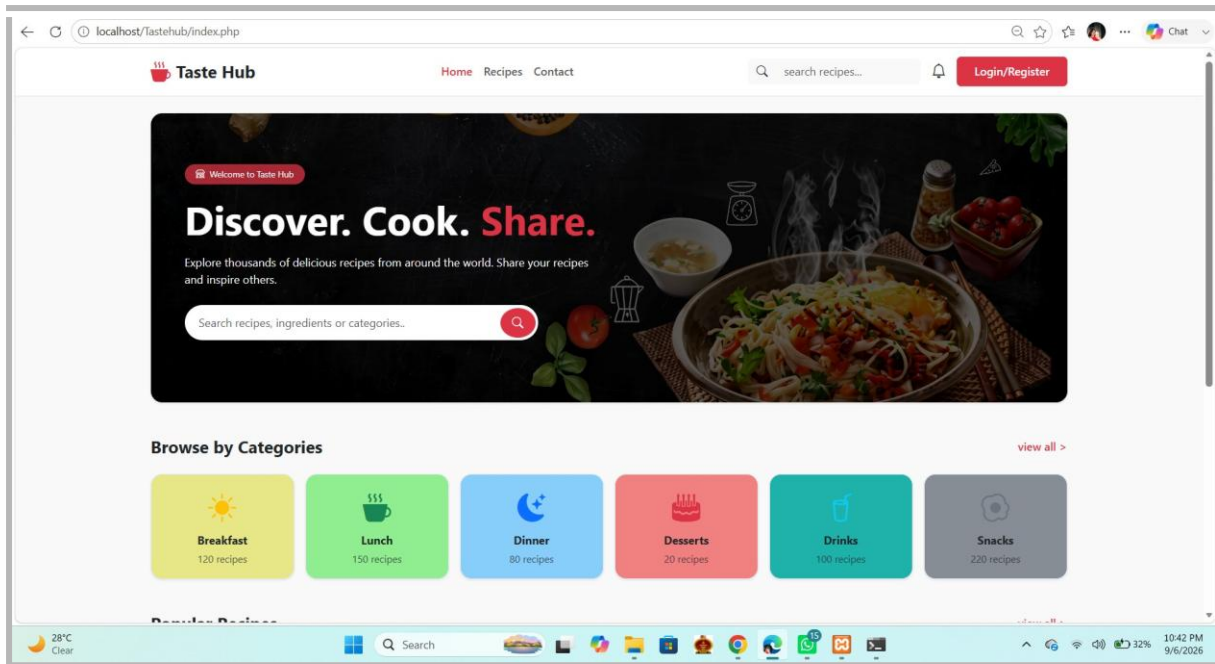
The actual project structure adheres to the prescribed course layout. All core modules are neatly decoupled: client-side assets ([css/](#), [js/](#), [images/](#)), database configurations ([includes/](#)), authentication workflows ([auth/](#)), and root-level application endpoints ([index.php](#), [recipes.php](#), [recipe-details.php](#), [contact.php](#), and [dashboard.php](#)).

3. Implementation

3.1 User Authentication (Register / Login / Logout)

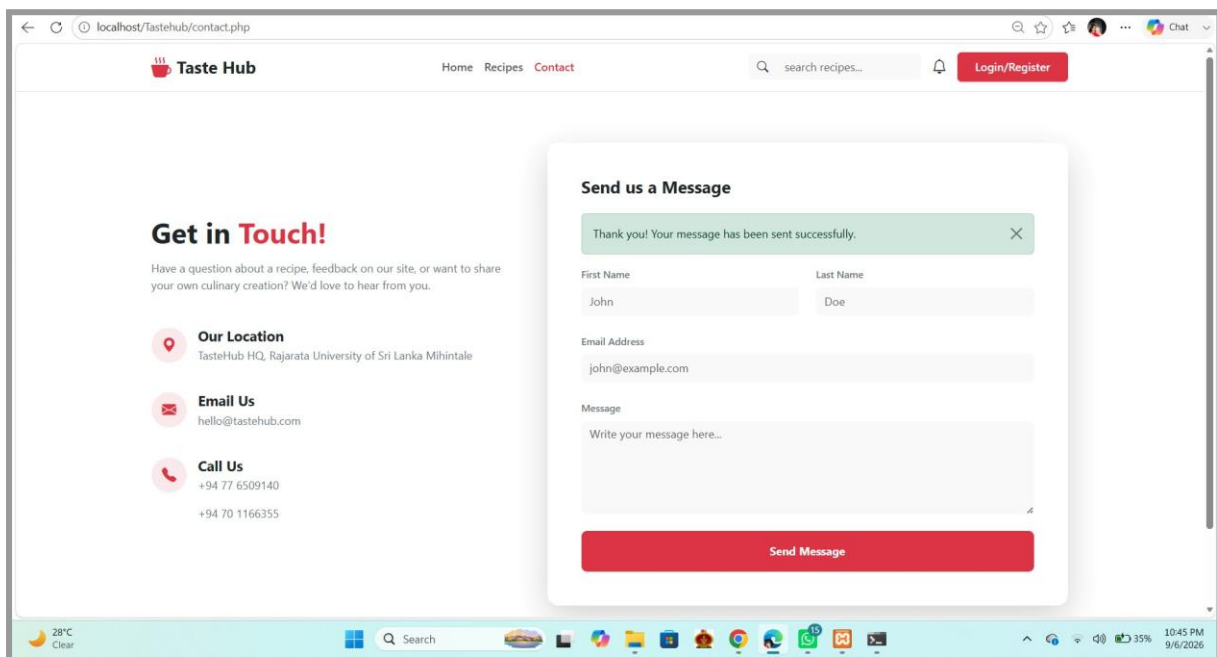
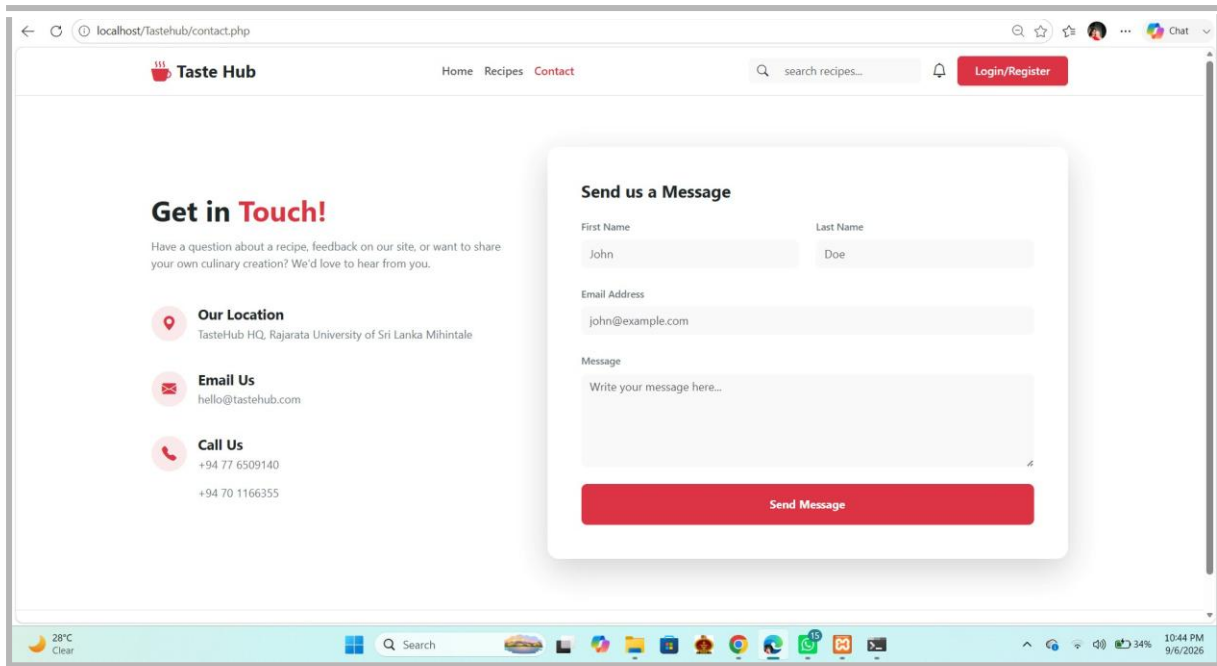
We implemented a secure authentication flow. Registration validates inputs and hashes passwords using `PASSWORD_BCRYPT`. The login script verifies the hash, starts a session, calls `session_regenerate_id()` to prevent session fixation, and routes the user to `dashboard.php`. Logging out securely destroys the session





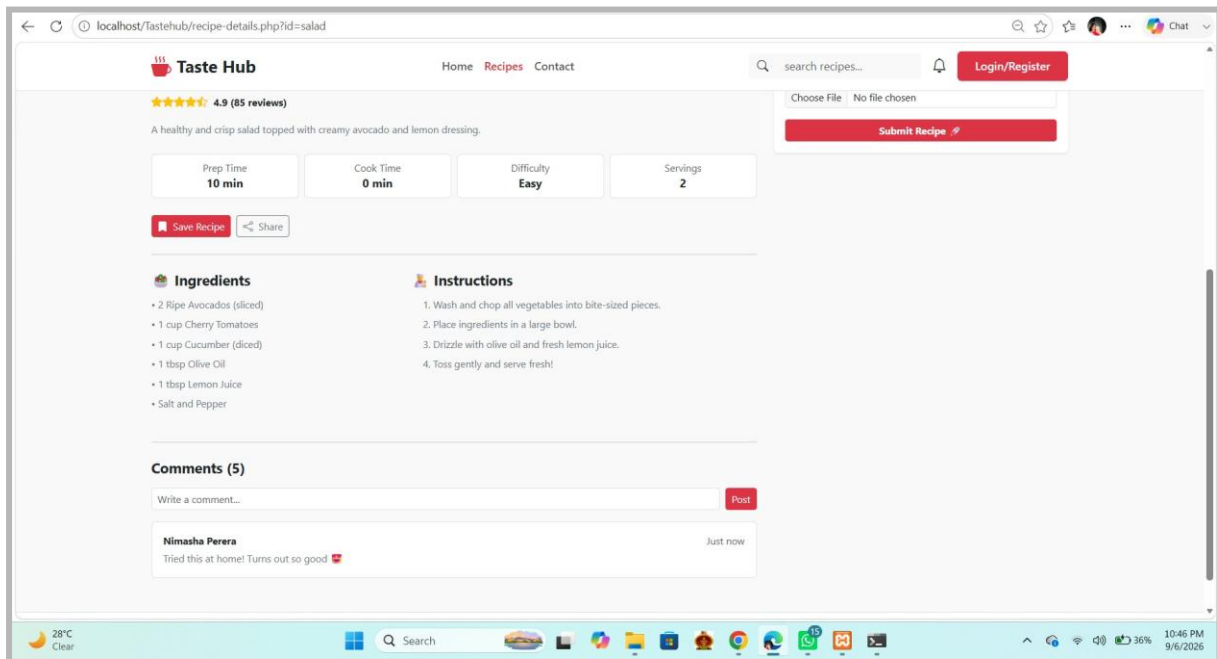
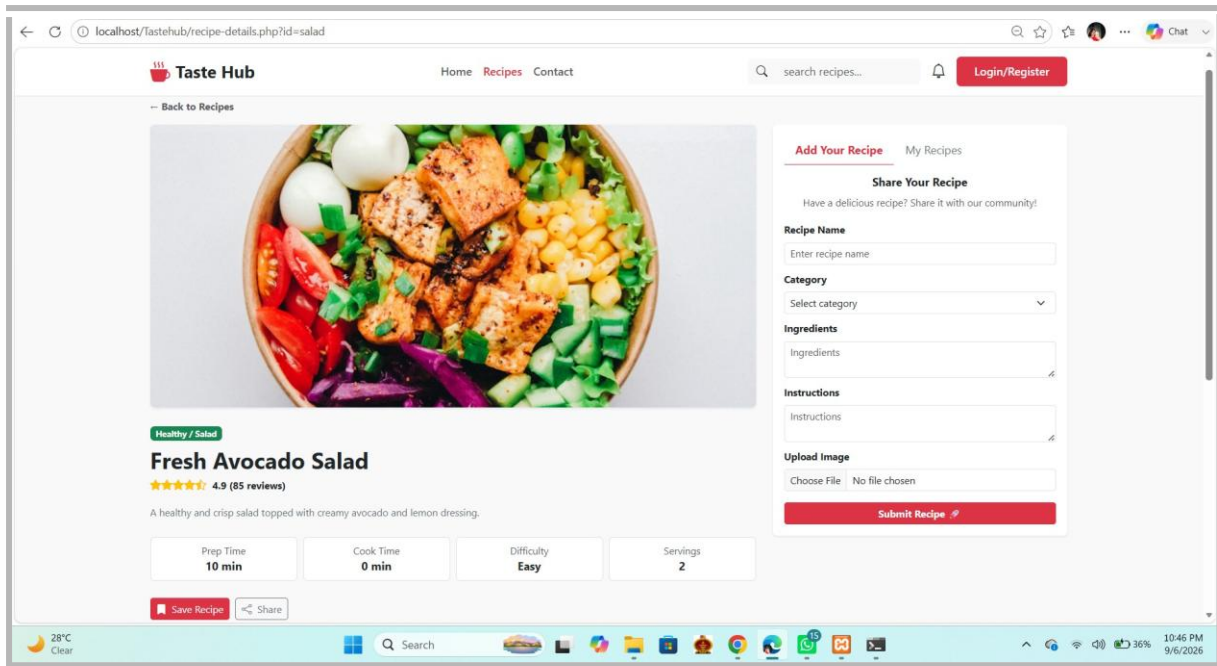
3.2 Contact Form

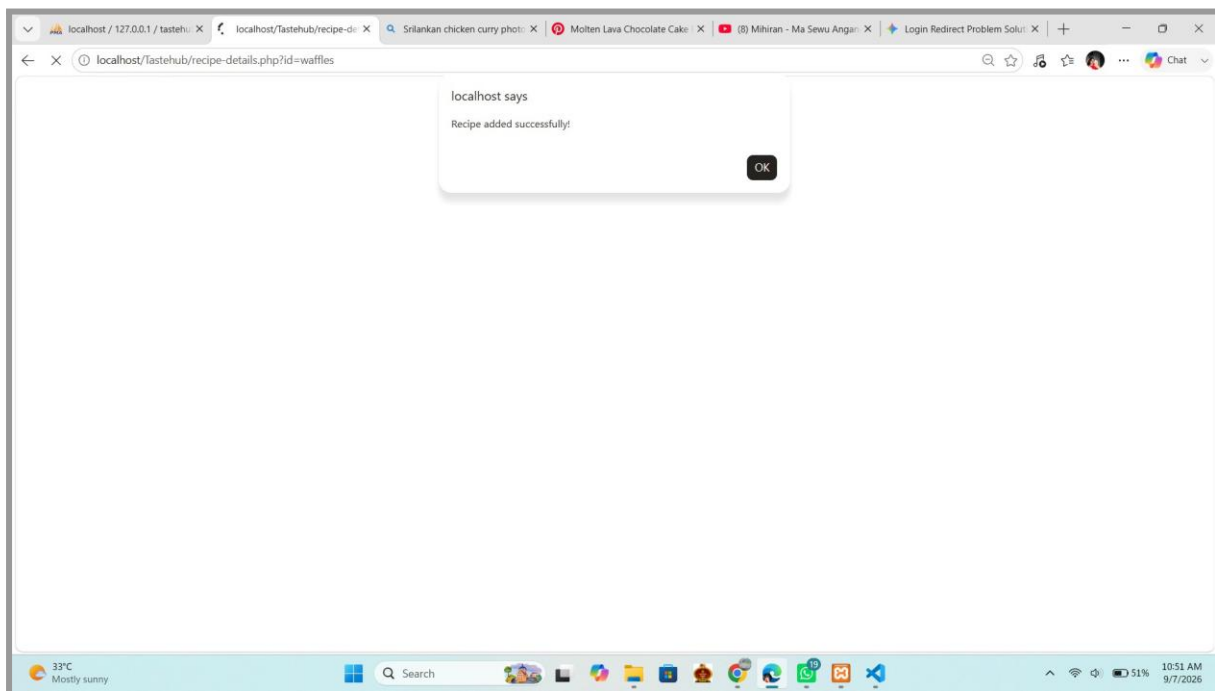
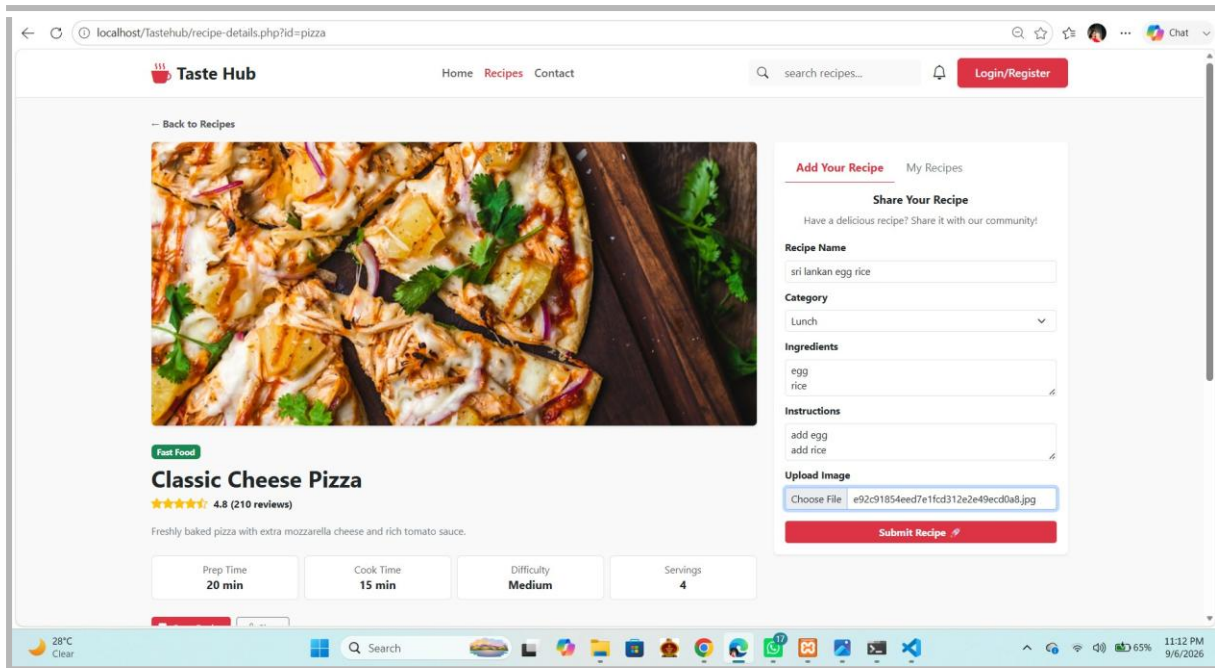
The contact page ([contact.php](#)) features a responsive form that captures user feedback. Upon submission, PHP sanitizes the inputs and executes a PDO prepared statement to insert the data into the [messages](#) table, followed by displaying a dynamic Bootstrap success alert.



3.3 Recipe Search Submission

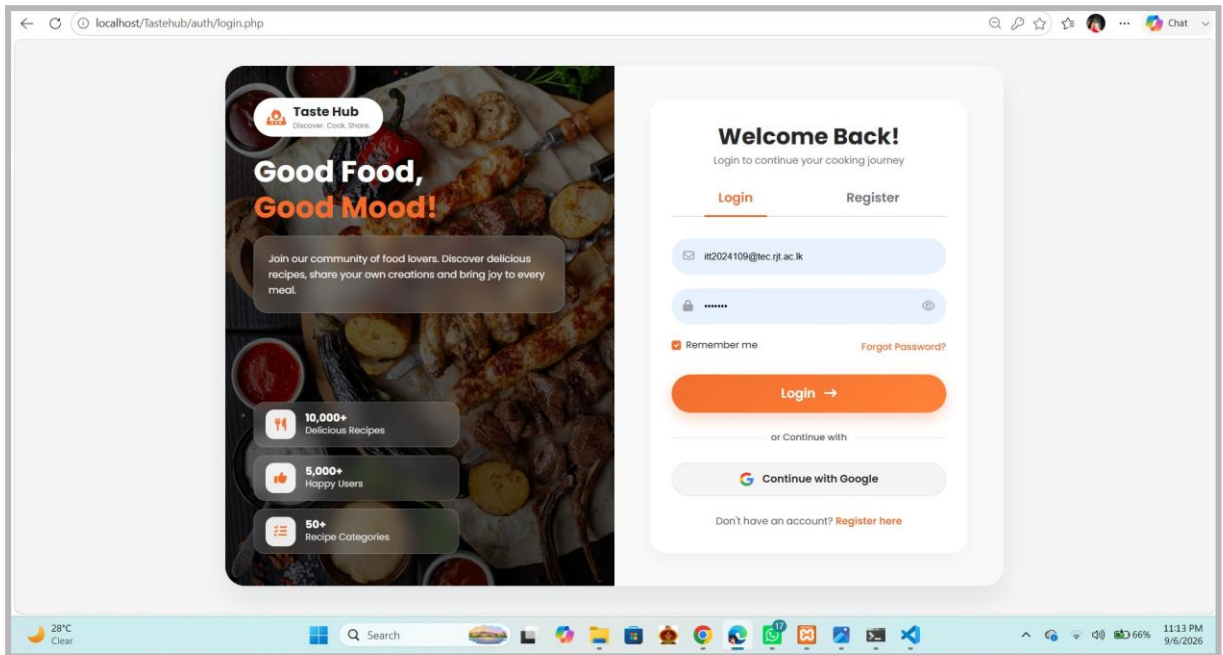
The `recipe-details.php` page is the core of TasteHub. It dynamically renders recipe data (prep time, ingredients, instructions). It also includes a robust form allowing users to submit new recipes, utilizing PHP's `move_uploaded_file()` with a unique `time()` timestamp to handle image uploads securely into the `images/` directory



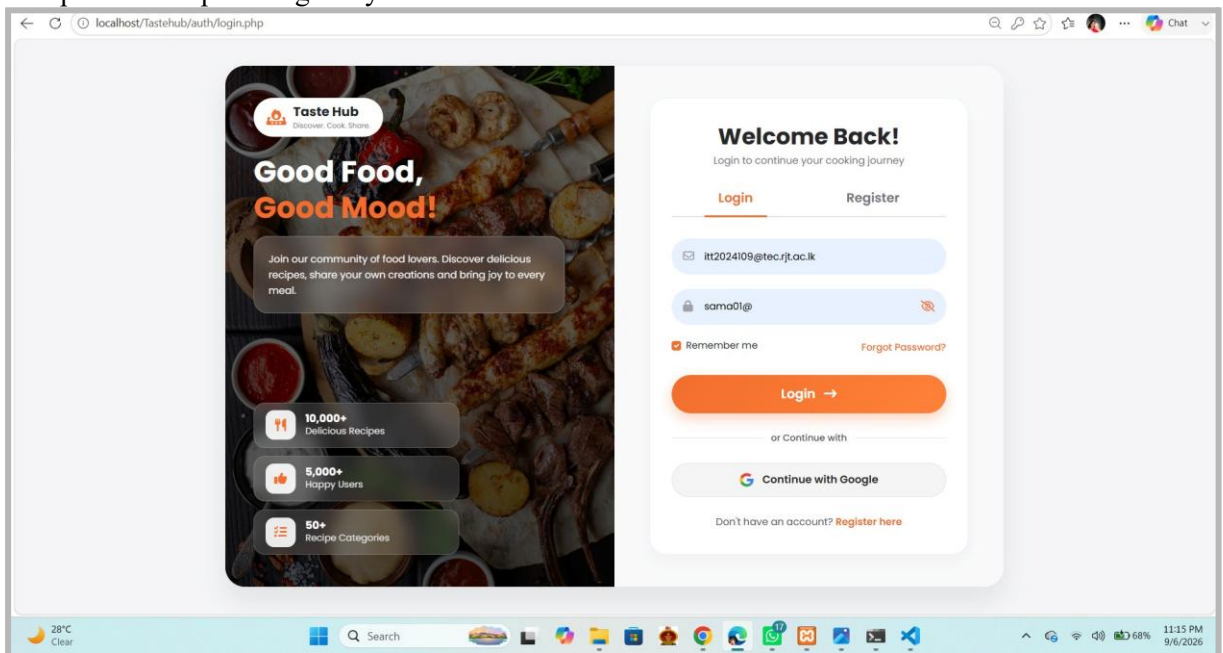


3.4 JavaScript Features

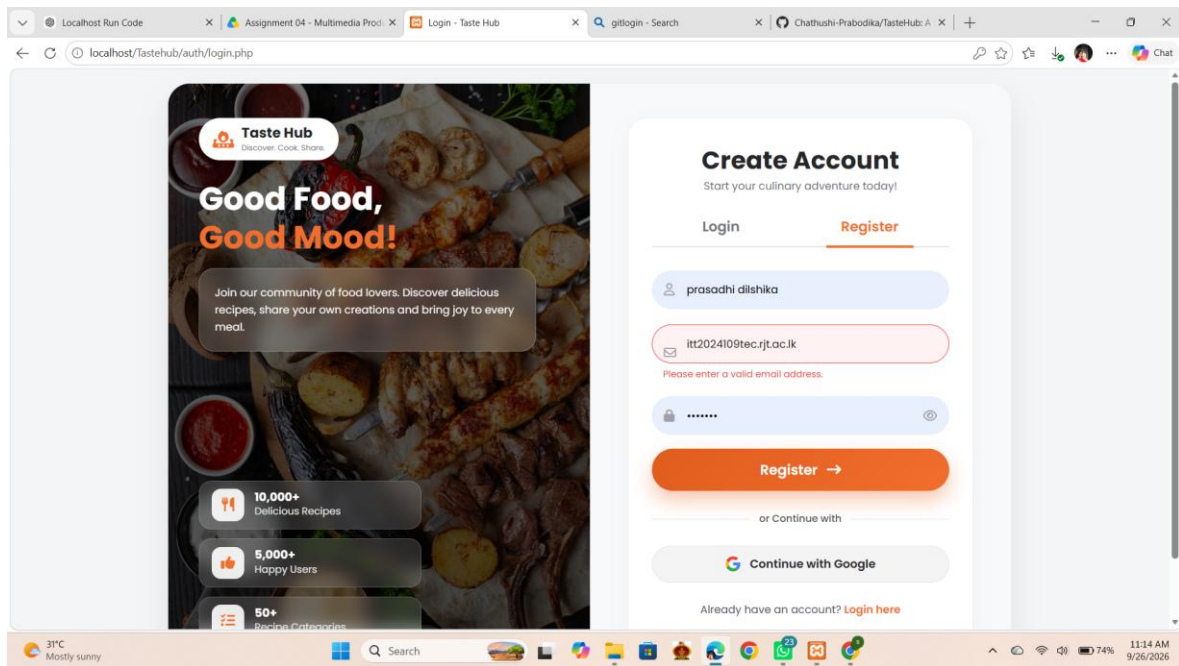
- **Dynamic Tab Switcher:** A custom `switchTab()` function seamlessly toggles between Login and Register views within a single interface without full page reloads.



- **Password Visibility Toggle:** Interactive DOM manipulation allowing users to show or hide their password input using an eye icon.



- **Form Validation:** Robust client-side validation for email formats (RegExp) and minimum password length checks, utilizing custom styled error messages and `preventDefault()` logic before submission



3.5 Security Measures

- Passwords hashed using `password_hash()` with `PASSWORD_BCRYPT`.
- Prepared statements (PDO) used for all database inserts (Contact and Recipes) to prevent SQL injection.
- Session handling: `session_regenerate_id()` called after login.
- Input sanitization using `trim()` on all form submissions.
- Image upload security: Unique file renaming using `time()` to prevent file overwriting

4. Challenges and Solutions

Challenge	Solution
Securely handling user image uploads without overwriting existing files with the same name.	Implemented a PHP script using the <code>time()</code> function concatenated with <code>basename()</code> to generate a unique filename for every upload

Providing a smooth user experience on the authentication page without loading separate HTML files.	Built a dynamic tab switcher using Vanilla JavaScript to toggle CSS visibility classes, creating a modern single-page application feel
--	--

5. Individual Contribution

Member	Tasks / Sections Owned	% Contribution
Chathushi Prabodika	Home Page, Login/ Register Page/Register Page , Java script feature1	50%
Samadhi Asanthika	Recipe details & Add page, Css styling & styling uniformity, Java script feature 2 and 3	50%

6. Setup Instructions

Steps to run the project locally using XAMPP/WAMP:

- Install XAMPP and start the **Apache** and **MySQL** modules.
- Copy the **TasteHub** project folder into the **htdocs/** directory (**C:\xampp\htdocs**).
- Open phpMyAdmin (<http://localhost/phpmyadmin>) and create a new database named **TasteHub_db**.
- Import the **database.sql** file into the new database.
- Update database credentials in **includes/db.php** if necessary.
- Visit <http://localhost/TasteHub/index.php> in the browser.

7. References

- Bootstrap Core Team. (2023). *Bootstrap 5.3 Documentation*. Retrieved from <https://getbootstrap.com/>

- PHP Group. (2024). *PHP: Hypertext Preprocessor Documentation*. Retrieved from <https://www.php.net/docs.php>
- Mozilla Developer Network (MDN). (2023). *JavaScript Guide*. Retrieved from <https://developer.mozilla.org/>